

RQ2 - Source and Platform Bias in GlobalGeoTree

How data sources shape geographic and taxonomic patterns

Research Question

RQ2 - Source and Platform Bias

To what extent do data sources shape the observed geographic and taxonomic patterns in **GlobalGeoTree**?

This report asks:

1. Which sources contribute the largest share of observations?
2. Is the dataset dominated by citizen-science platforms such as iNaturalist?
3. Do different sources cover different countries or regions?
4. Are some sources associated with specific years, countries, or species groups?

The analysis is descriptive. Strong source-specific patterns should be interpreted as evidence of **recording and platform bias**, not only as biological patterns in tree distributions.

Data Loading

Measure	Value
Observation file	GlobalGeoTree.csv
Rows analysed	6,263,345
Distinct sources	2,493
Distinct source groups	6
Distinct countries	221
Distinct species	21,001
Year column available	Yes
Coordinate columns available	Yes
Category source	GlobalGeoTree-10kEval-900.csv joined by species_key
Observations with known category	2,205,871
Species with known category	941

This report uses 6,263,345 observations.

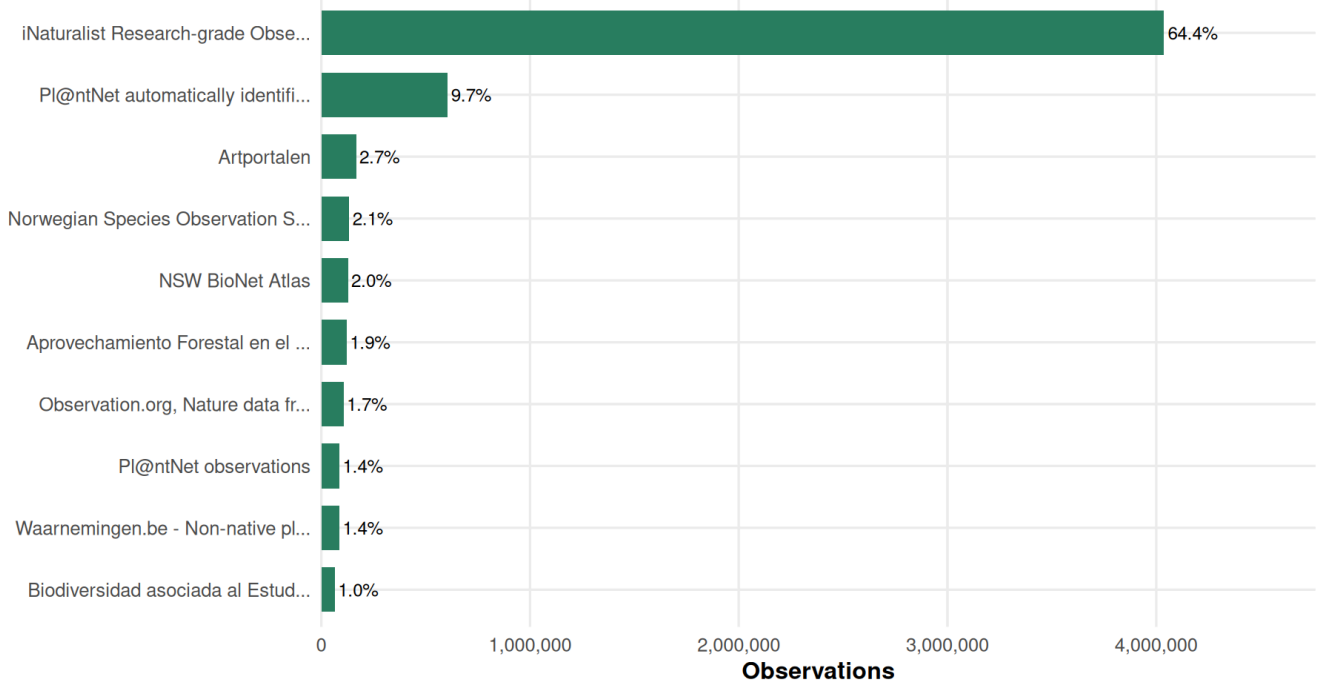
Headline Result

The source structure is highly concentrated: **iNaturalist alone contributes 64.5%** of observations, while **iNaturalist and Pl@ntNet together contribute 75.5%**. Therefore, many observed geographic, temporal, and taxonomic patterns should be read partly as platform-coverage patterns.

Source Concentration

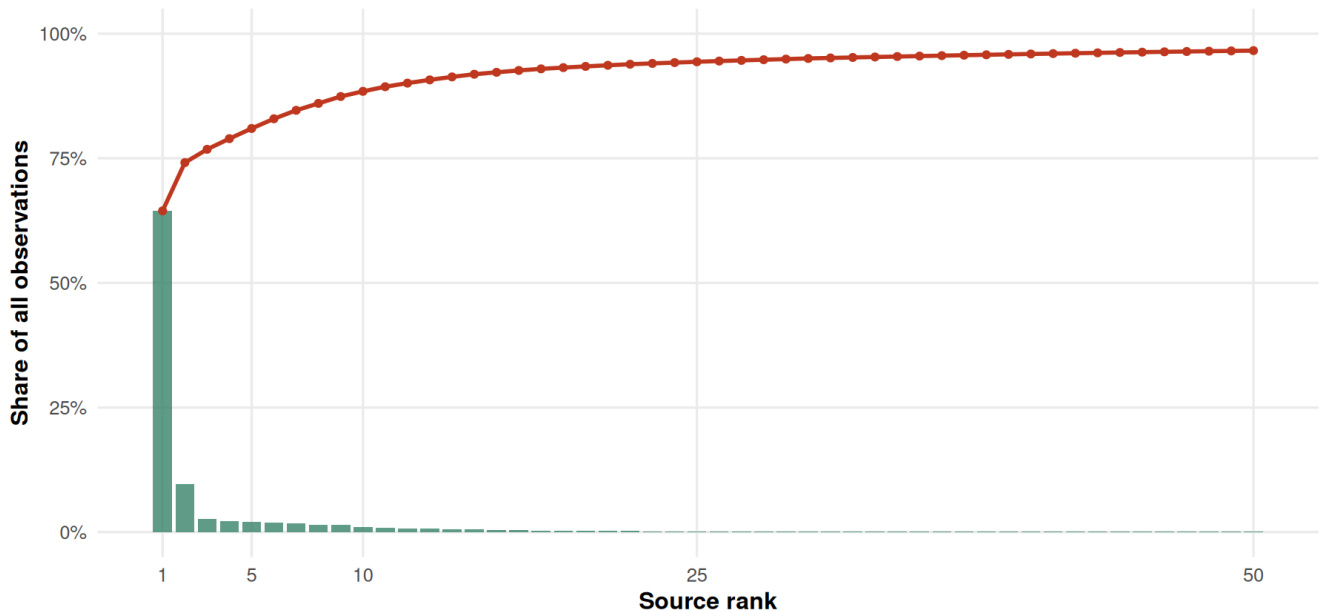
Largest observation sources in GlobalGeoTree

The top source is large enough to shape the overall dataset.



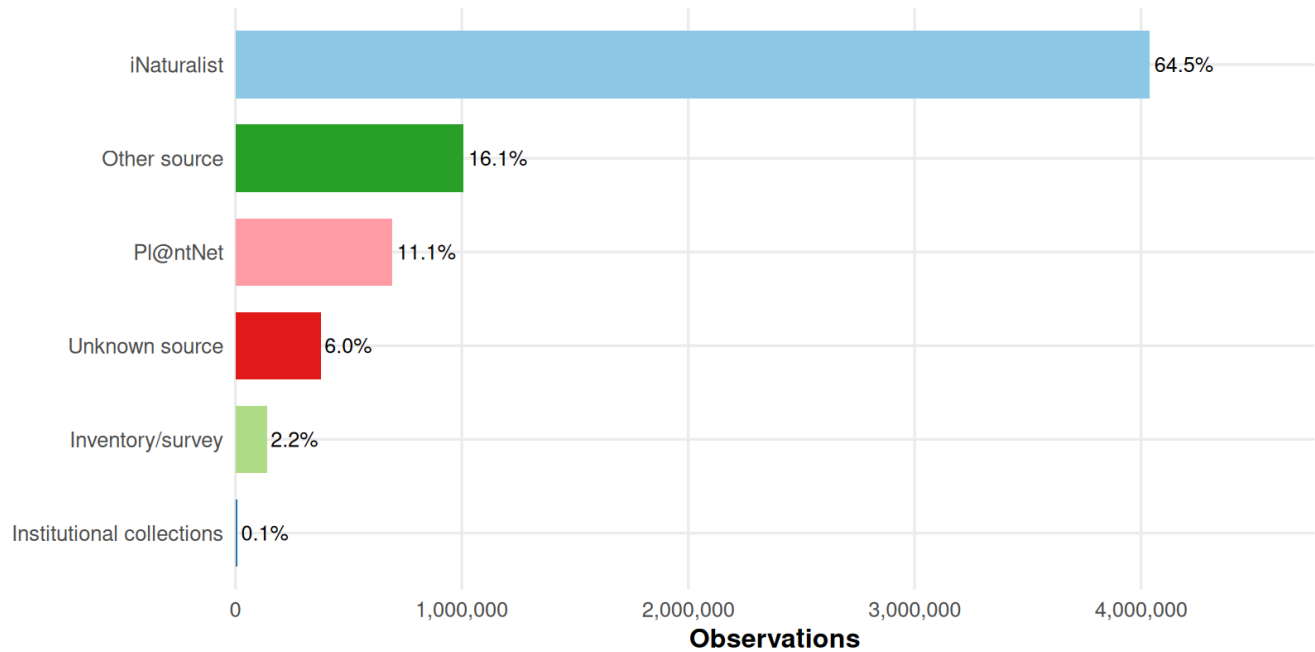
A small number of sources explain most observations

Bars show each source's individual share; the red line shows cumulative share.



Observation share by broad source group

Citizen-science platforms dominate the grouped source structure.



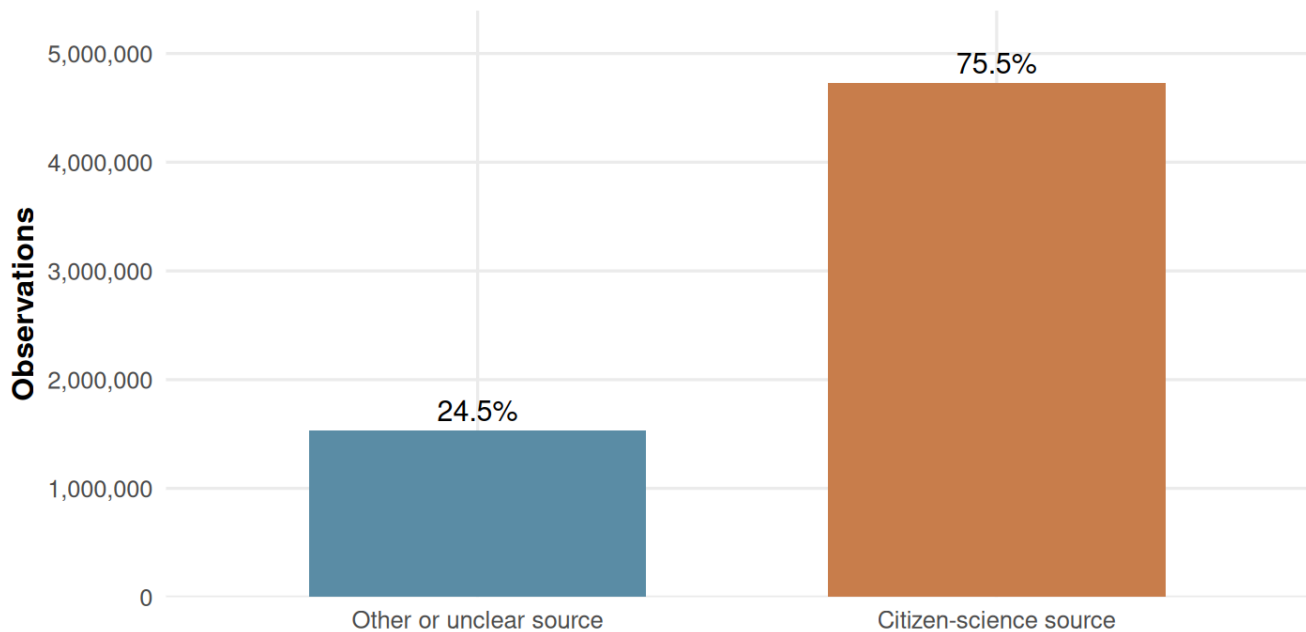
Source	Observations	Share	Cumulative share
iNaturalist Research-grade Observations	4,036,465	64.4%	64.4%
Pl@ntNet automatically identified occurrences	606,258	9.7%	74.1%
Artportalen	166,929	2.7%	76.8%
Norwegian Species Observation Service	134,097	2.1%	78.9%
NSW BioNet Atlas	128,009	2.0%	81.0%
Aprovechamiento Forestal en el área de influencia del Proyecto Hidroeléctrico Ituango	121,900	1.9%	82.9%
Observation.org, Nature data from around the World	106,352	1.7%	84.6%
Pl@ntNet observations	86,806	1.4%	86.0%
Waarnemingen.be - Non-native plant occurrences in Flanders and the Brussels Capital Region, Belgium	86,599	1.4%	87.4%
Biodiversidad asociada al Estudio de Impacto Ambiental del proyecto: Interconexión Cuestecitas - Copey - Fundación 500/220 mil voltios	65,000	1.0%	88.4%

Interpretation. The distribution is not balanced across sources. A few large platforms dominate the evidence base, so global summaries are strongly affected by the behaviour, geography, and taxonomy covered by those platforms.

Citizen-Science Dominance

Citizen-science platforms dominate the observations

Citizen-science is defined here as iNaturalist plus Pl@ntNet.

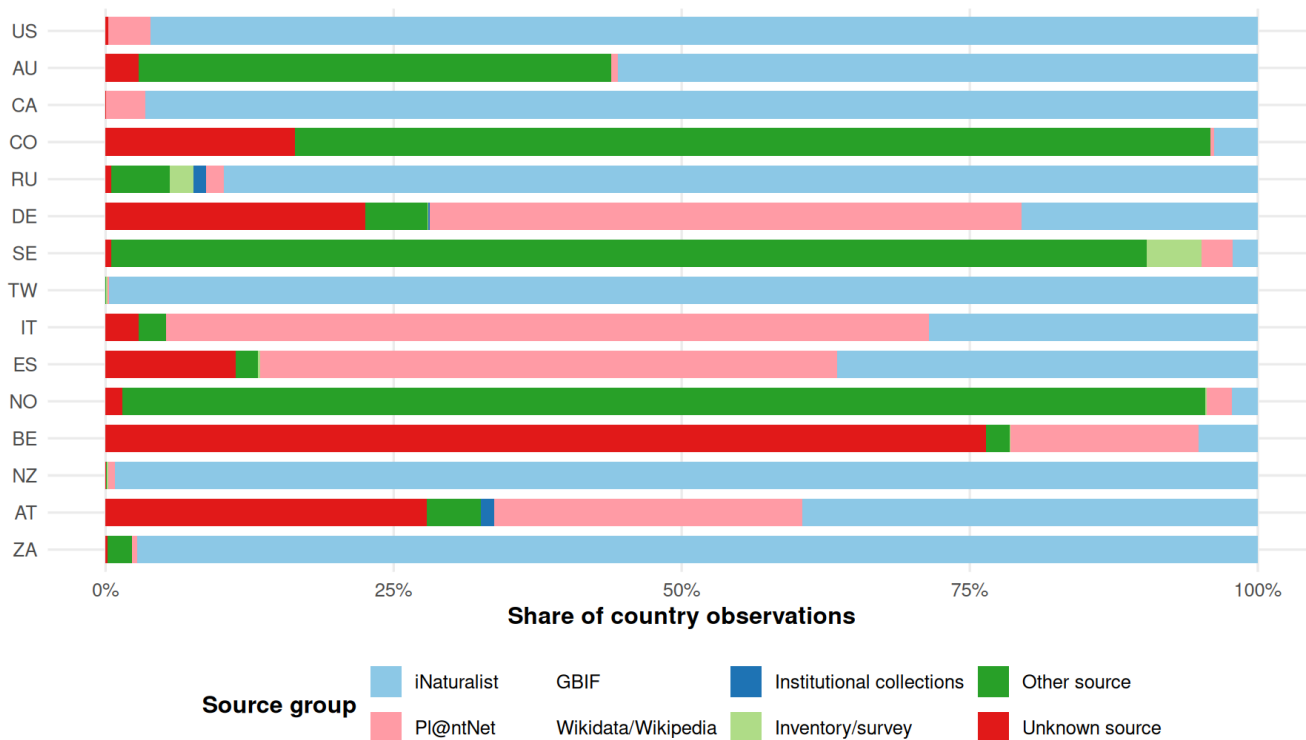


Interpretation. Citizen-science dominance does not make the data unusable, but it changes how the data should be interpreted. Places and species that are popular with users or easy to photograph may be overrepresented.

Countries and Regional Coverage

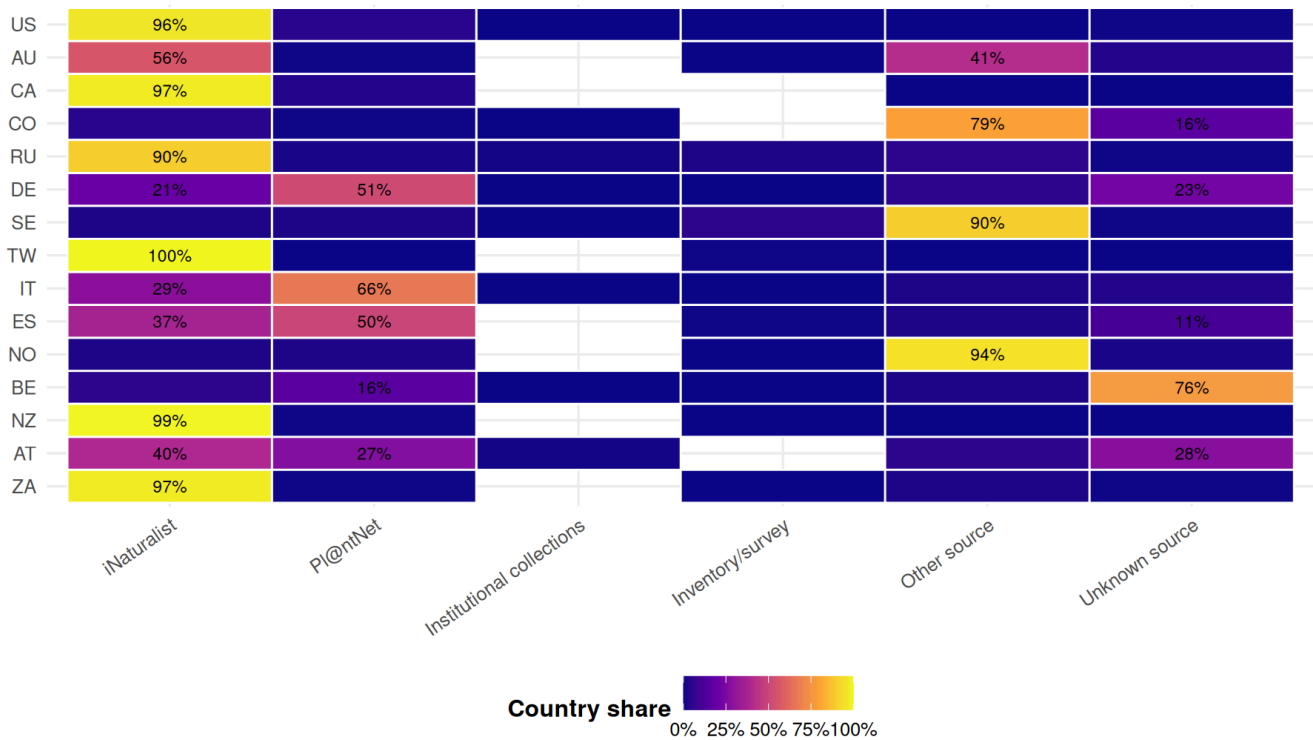
Source composition differs by country

Top 15 countries by observations; bars sum to 100% within each country.



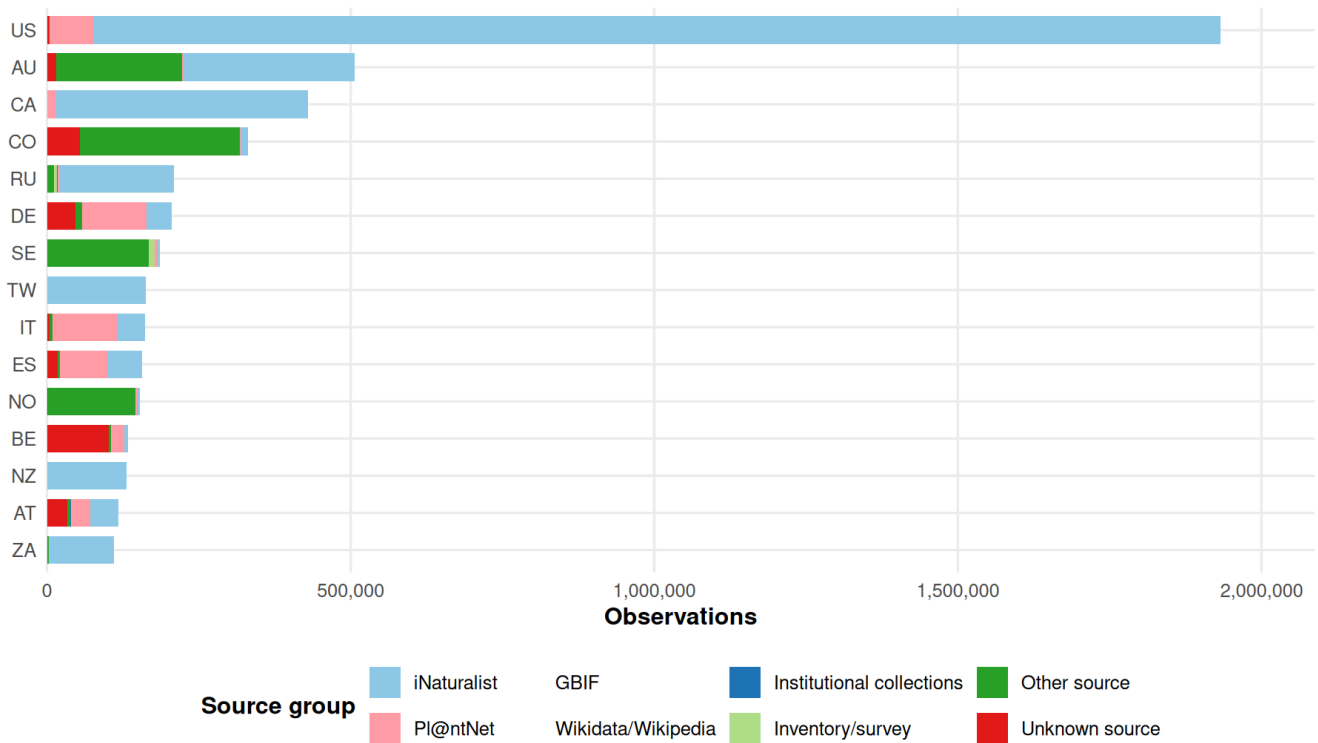
Country-by-source heatmap

Darker cells show countries that rely heavily on one source group.



Observation volume by country and source group

Country comparisons combine source composition and very unequal record volumes.

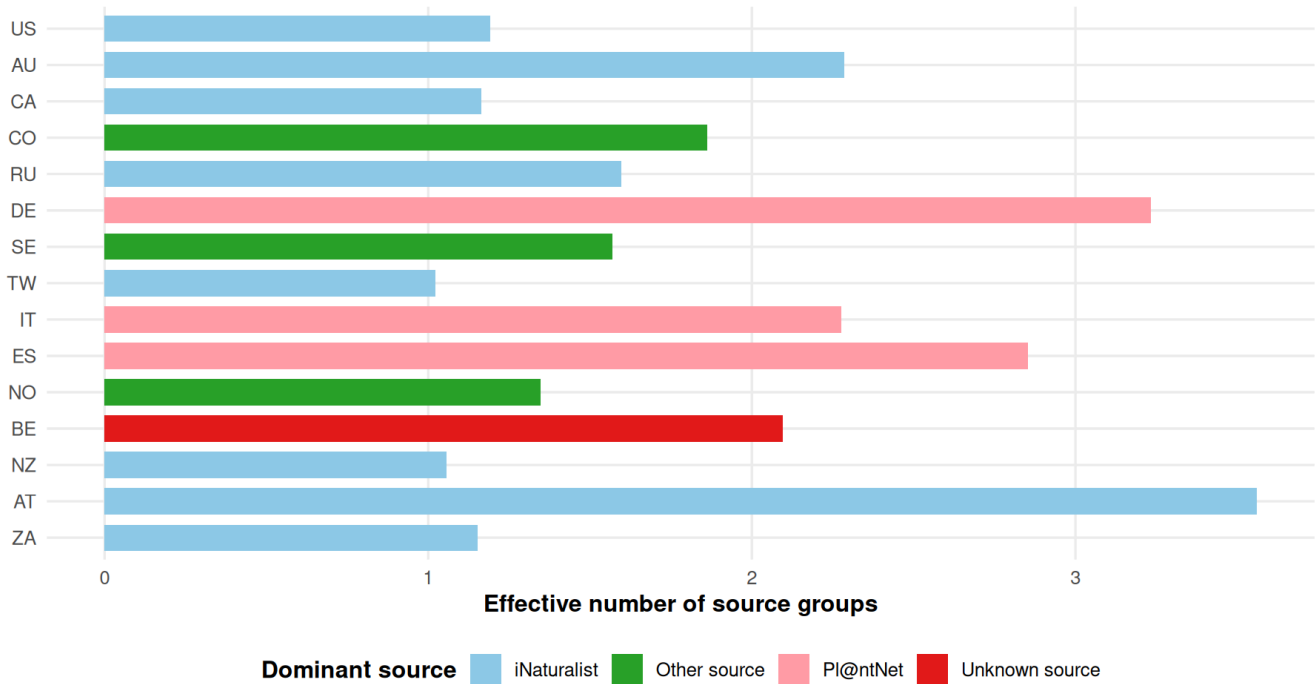


Interpretation. Countries differ both in the number of observations and in the source mix. This means that geographic comparisons can be biased by platform adoption. A country dominated by iNaturalist is not directly comparable to a country dominated by local inventories or unknown sources.

Source Diversity by Country

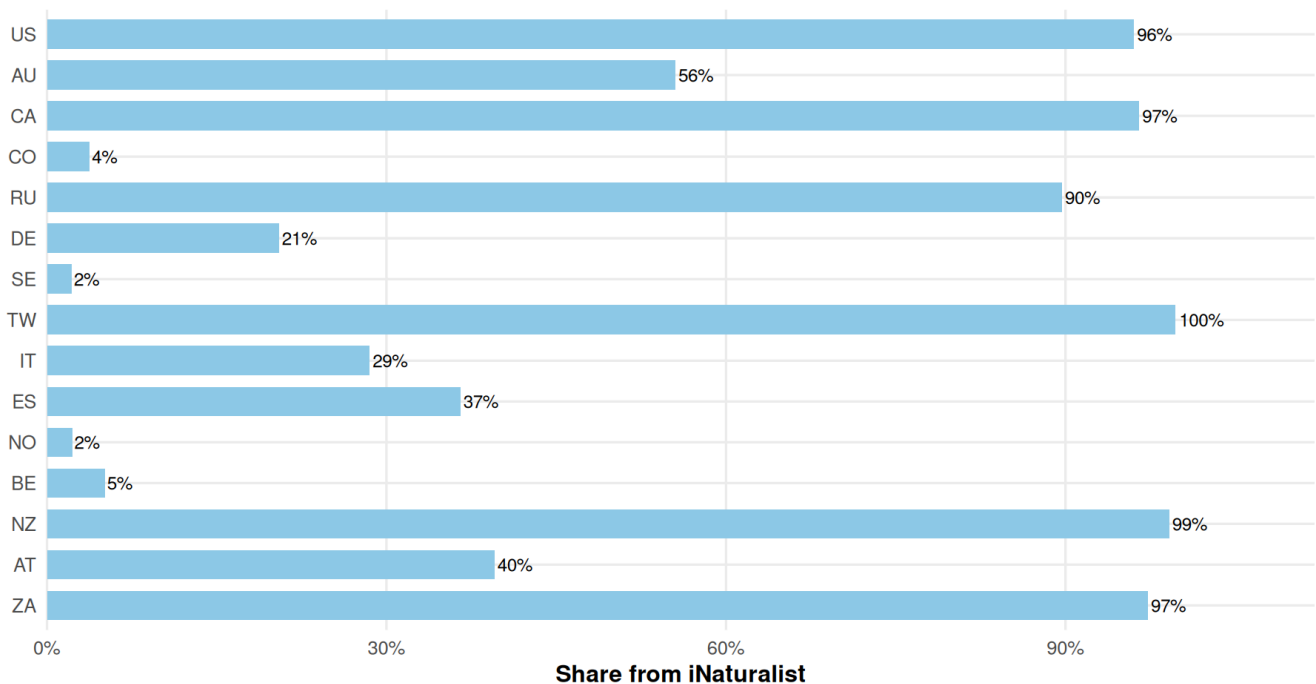
How source-diverse are the largest country samples?

Effective number of source groups is low when one source dominates a country.



iNaturalist share by country

Top 15 countries by observation count.

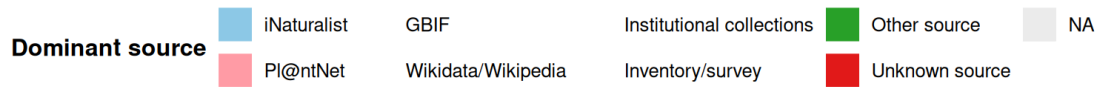
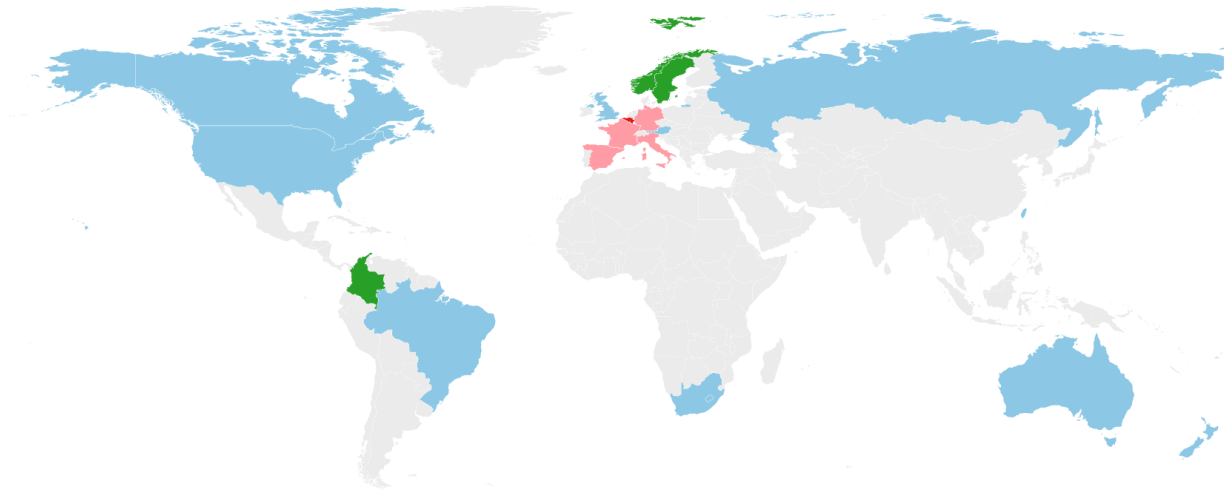


Interpretation. Source diversity provides a compact way to see which countries depend on one platform and which countries combine several source groups. Low diversity means higher risk that country-level patterns are platform-specific.

Optional Map: Dominant Source by Country

Dominant source group by country

Countries are colored by the source group contributing the most observations.

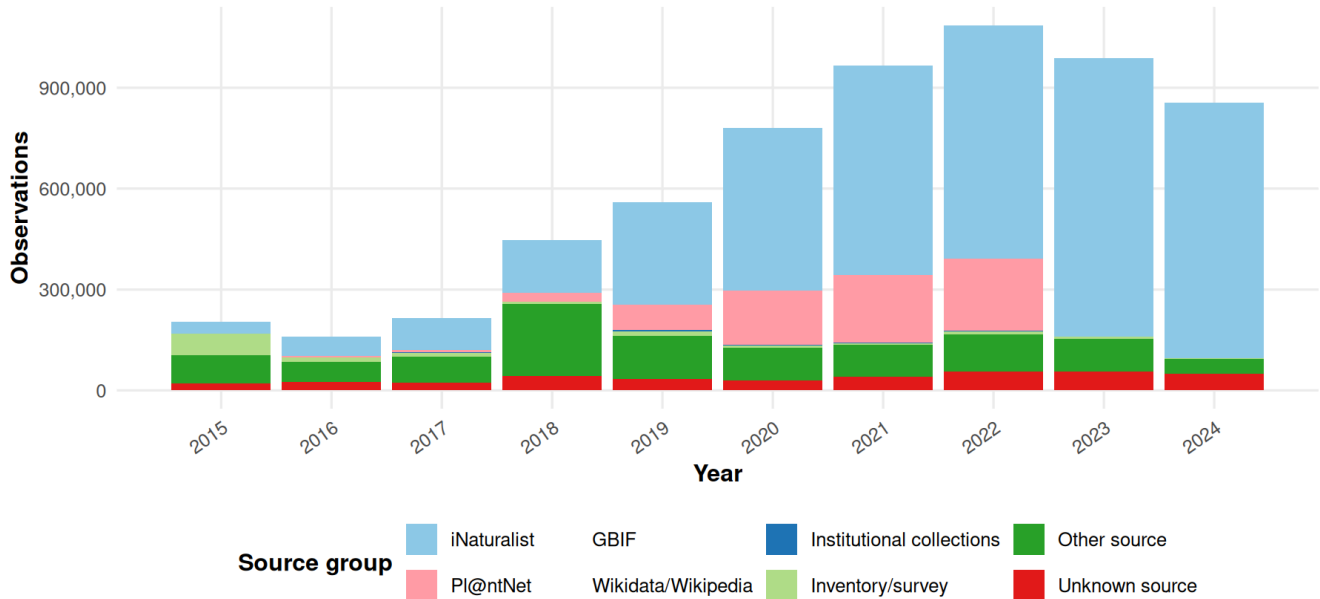


This optional map uses the maps package and a small built-in country-name lookup.

Sources Over Time

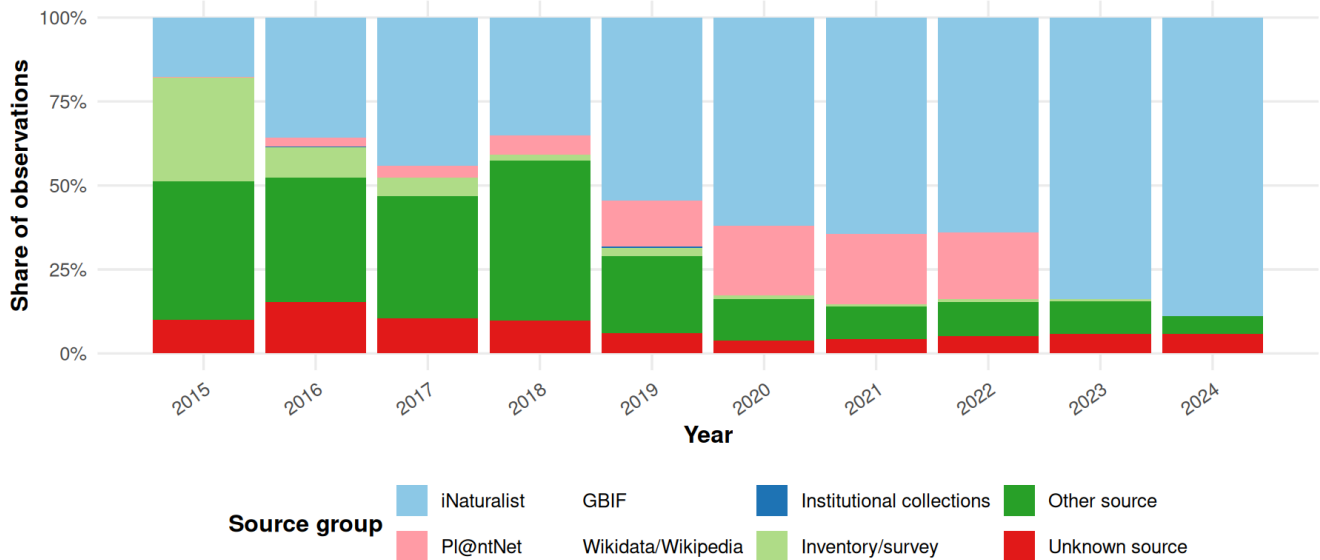
Observation sources by year

The rise of a platform can look like a temporal trend in tree observations.



Source composition by year

Bars sum to 100% within each year, revealing shifts in source dominance.



The most recent year may be incomplete depending on the data export date.

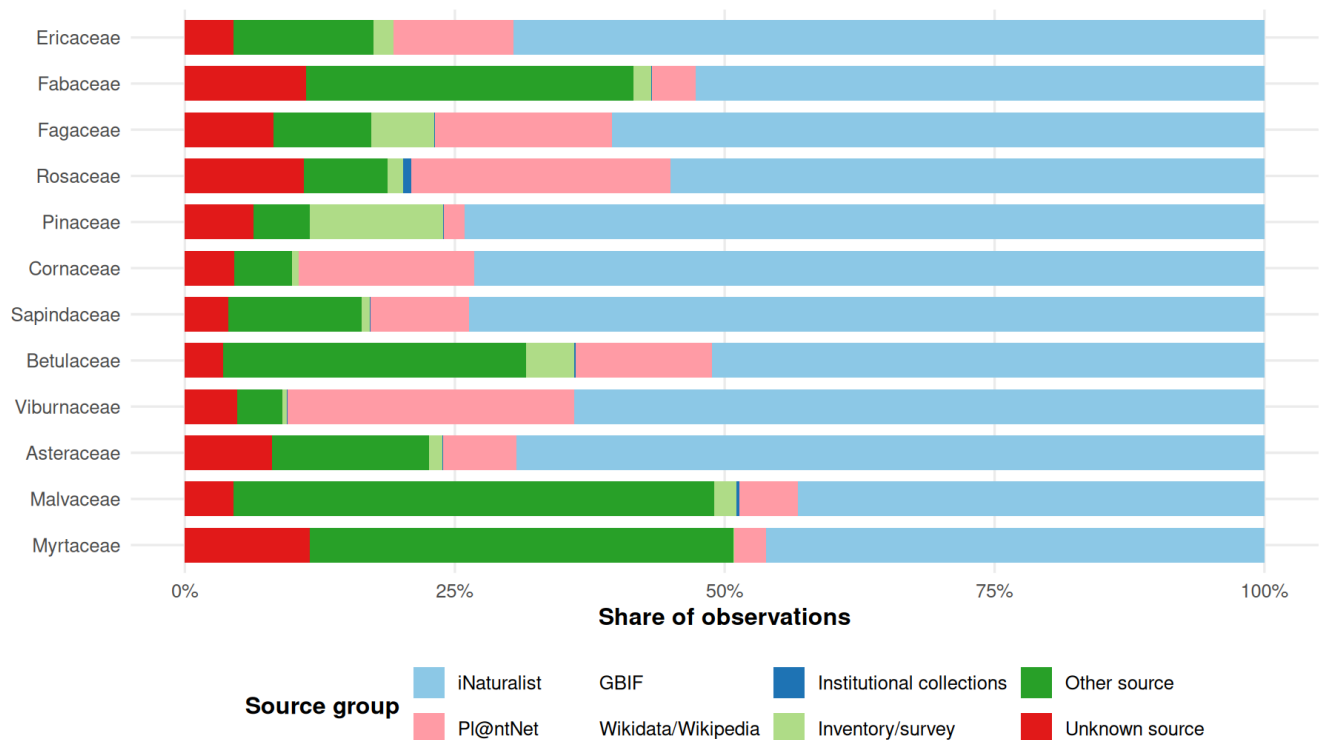
Interpretation. Temporal patterns are not source-neutral. If iNaturalist or PI@ntNet grows in recent years, the dataset may show an apparent rise in observations even if actual tree occurrence is unchanged.

Sources and Taxonomic Groups

This section checks whether source groups disproportionately record particular families, genera, or species. Family is used first when available because it is more readable than hundreds of species names.

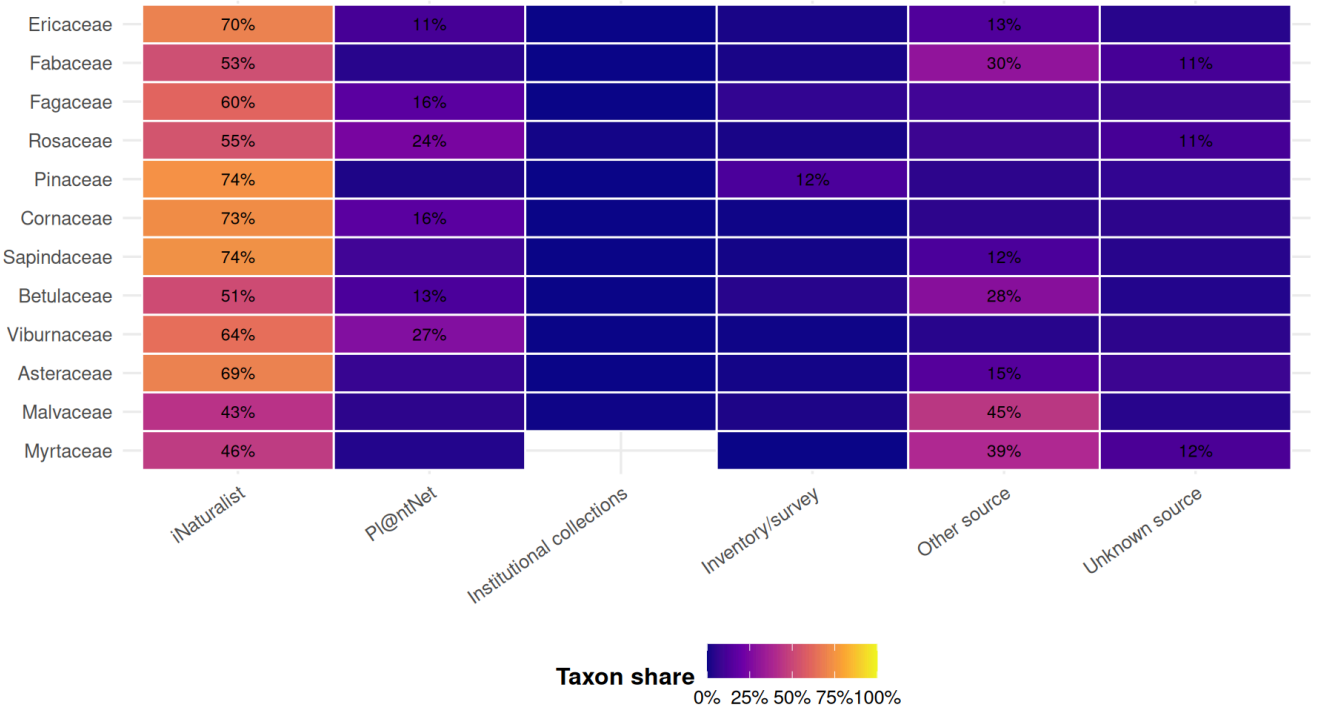
Source composition by family

Bars sum to 100% within each taxonomic group.



Taxonomic groups are not equally represented across sources

Heatmap by family and source group.

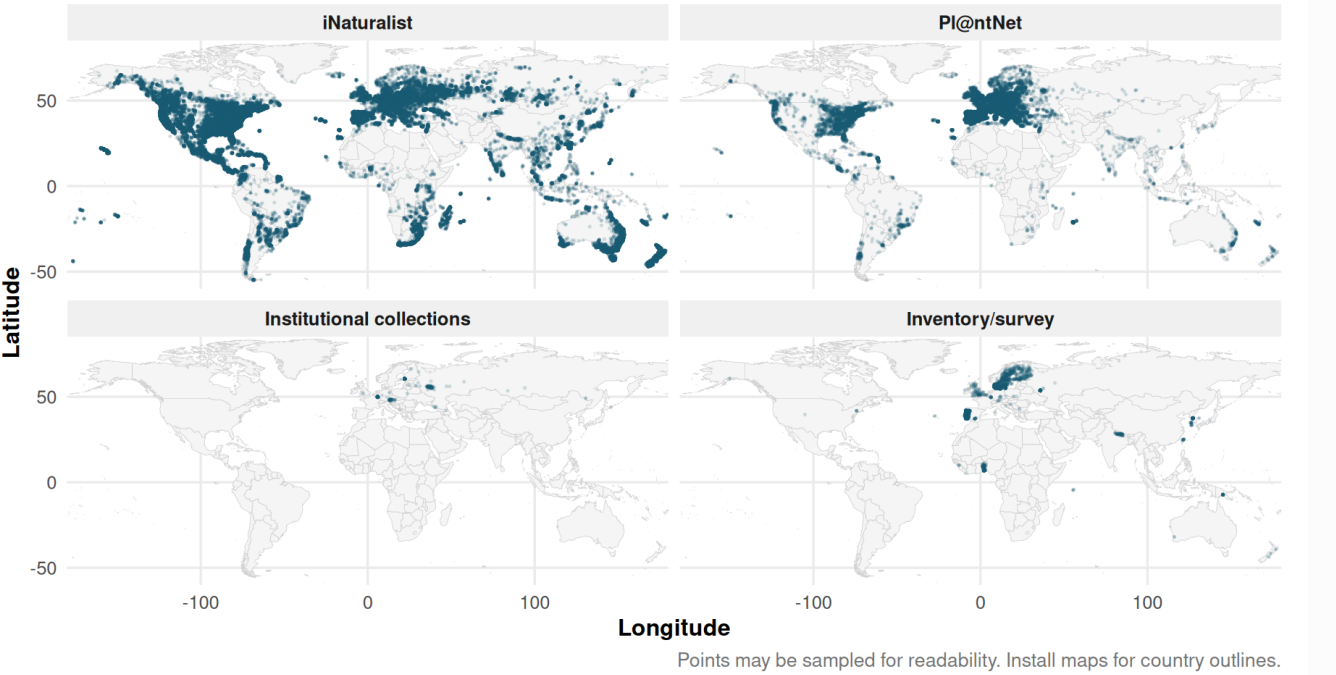


Interpretation. If a family or genus is mostly recorded by one platform, its geographic pattern may partly reflect that platform’s user base and coverage.

Geographic Footprints of Major Sources

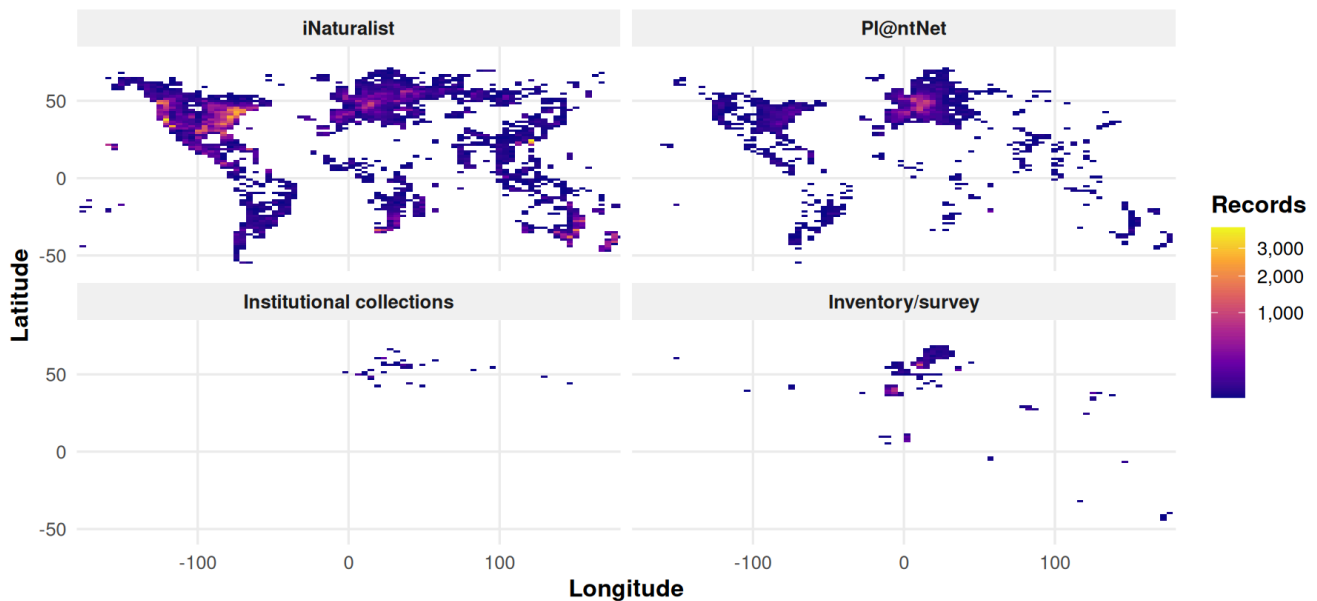
Geographic coverage of major source groups

Different platforms leave visibly different spatial footprints.



Observation density by source group

Binned counts reveal dense recording regions more clearly than raw points.

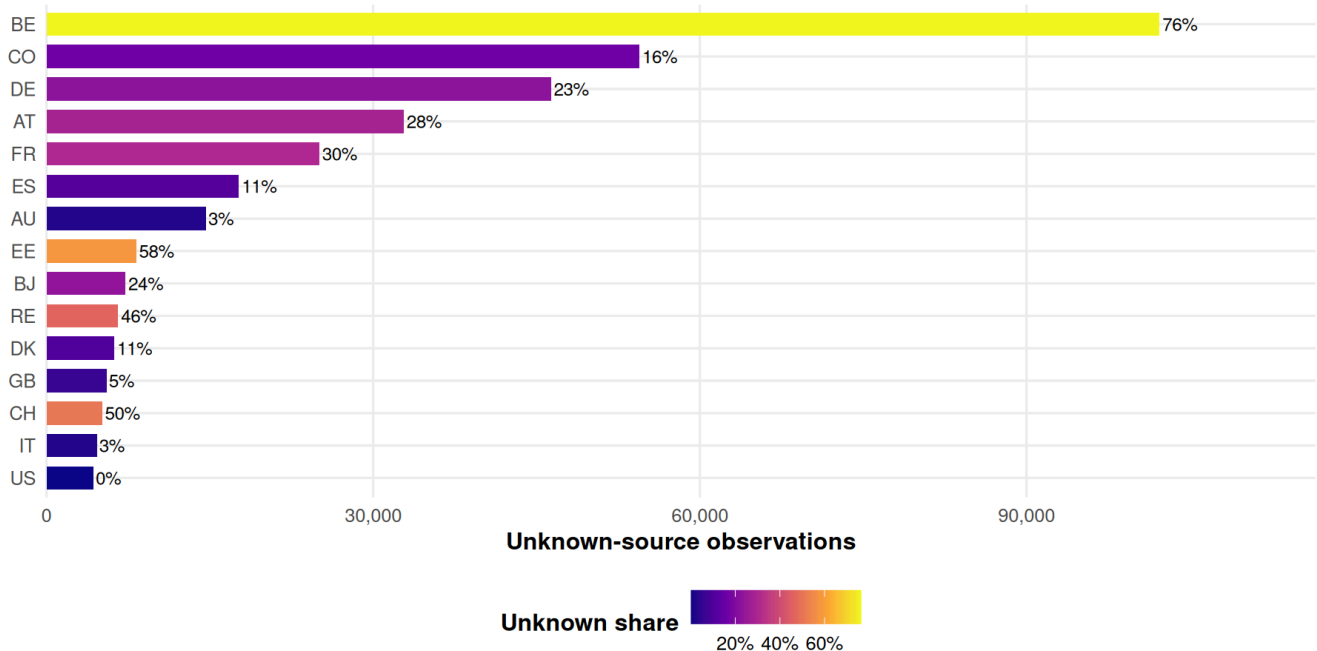


Interpretation. The source maps show that platform coverage is geographically uneven. Some regions are visible mainly because they are covered by a particular platform or source pipeline.

Unknown Source Diagnostics

Where are unknown-source observations concentrated?

Countries with many unknown-source records need cautious interpretation.

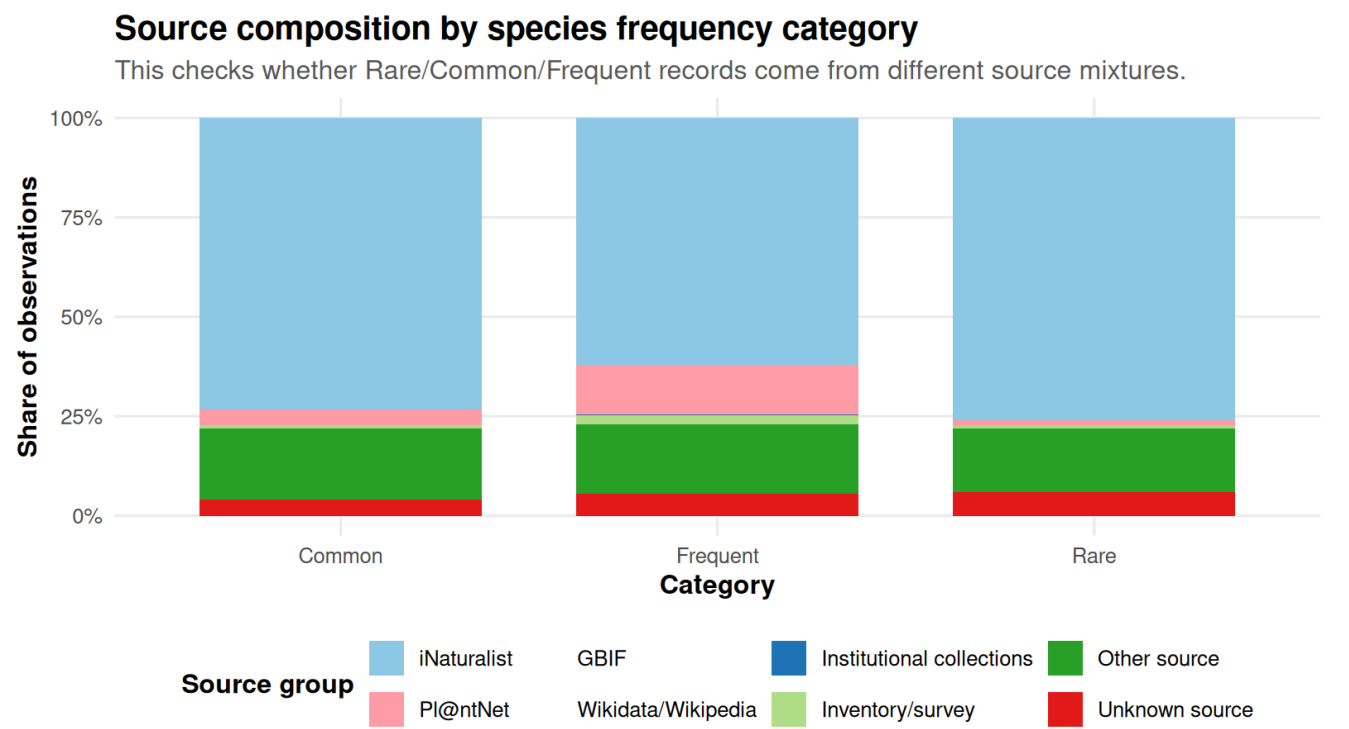


Interpretation. Unknown source records are a metadata-quality issue. If they are concentrated in particular countries, those countries have weaker evidence for source-specific interpretation.

Rare, Common, and Frequent Categories

This section is shown only when category labels are available either inside **GlobalGeoTree** or through the optional eval lookup file.

Measure	Value
Category source	GlobalGeoTree-10kEval-900.csv joined by species_key
Observations with known category	2,205,871
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Overall Interpretation

GlobalGeoTree is not source-neutral. The dataset contains millions of observations from thousands of named sources, but the observation structure is dominated by a small number of platforms, especially iNaturalist and Pl@ntNet. Source composition also varies by country, year, and taxonomic group. Therefore, observed geographic and taxonomic patterns should be interpreted as a combination of tree occurrence, recording effort, platform popularity, and source-specific data integration.

For downstream analyses, consider reporting source composition next to any global, country-level, temporal, or taxonomic result. Where possible, repeat key analyses with and without the largest citizen-science source to test robustness.